Ugonna Victor Emeka-Inegbu

Mount Pleasant, MI 48858 | (708) 664-8707 | emeka1uv@cmich.edu
https://ugonna-portfolio | http://www.linkedin.com/in/ugoemeka | https://github.com/HackUgo

EDUCATION

Central Michigan University, Mt Pleasant, MI **Expected May 2027**
Bachelor of Science in Electrical Engineering
Dean's List-Fall 2023-Present | GPA: 3.7/4.0

RELEVANT SKILLS

- **Product & Iterative Design:** CAD modeling (Fusion 360), 3D printing (PLA), rapid prototyping, mechanical assembly, system integration — applied across multiple hardware projects from concept to functional prototype.
- **Hands-On Fabrication:** PCB design (KiCad — schematic, layout, bring-up), hand-soldering (SMD/through-hole), component selection, thermal management, hardware validation.
- **Embedded Systems:** C/C++ firmware (ESP32/Arduino), PWM control, motor driver circuits, SPI/UART/I2C, sensor integration (IMUs, gyros).
- **Software & Simulation:** Python, MATLAB, LTspice, Git/GitHub, OpenCV, VS Code, Betaflight, ROS basics.
- **Testing & Validation:** Thermal testing, statistical quality analysis (variance, std dev), production validation.

PROFESSIONAL WORK EXPERIENC

Electrical Engineering Intern (Miller Electric, Appleton, WI) May 2025 - August 2025

PCB Redesign for Production

- Led redesign of welding machine output filter board after identifying thermal failure where resistors exceeded 180°C specification by 65°C during testing.
- Redesigned PCB layout in KiCad with optimized component spacing and upgraded to higher-wattage resistors; personally ordered components, hand-soldered final board, and validated 40% temperature reduction (245°C → 150°C) through thermal testing.

Inductor Quality Control Analysis Project

- Conducted quality analysis of 60 in-house manufactured inductors using the Bode 500 analyzer to measure leakage and magnetizing inductance.
- Performed statistical analysis (variance and standard deviation calculations) and identified a critical manufacturing outlier with significantly lower inductance, flagging a defect that surprised the senior engineer and prompted a manufacturing process review.

Additional Technical Experience: Validated circuit designs through LTSpice simulations and oscilloscope analysis; participated in board design review meetings and motor characterization testing using a dynamometer**.**

Product Design Instructor (Girls in Engineering Academy, CMU) July 2026-August 2026

- Designed and delivered 3-week Arduino-based underwater ROV curriculum for high school students across 4 teams, covering embedded systems, sensor integration, and motor control –
- Guided them through full engineering design cycle — including circuit assembly, programming, live ROV water deployment, and sample collection Beaver Island.

TECHNICALPROJECTS

ESP32-Controlled Robot Arm July 2026 - Present

- Developing a 5-DOF robot arm with ESP32 wireless control for real-time servo positioning and motion control
- 3D-printing a custom mechanical frame in PLA and integrating MG996R servo motors with PWM control.

Custom Quadrotor (Delivery drone) July 2026 – Present

- Engineered custom quadrotor from scratch, including precision soldering of ESCs, flight controllers, and motors.
- Performed PID tuning in Betaflight to optimize flight stability and controller responsiveness through iterative testing.
- Designing and 3D-printing a custom mechanical frame and payload release mechanism using Fusion 360.

Bluetooth RC Car (Breadboard to Custom PCB) May 2025 – August 2025

- Engineered a remote-controlled car with Arduino microcontroller and HC-05 Bluetooth module for wireless command processing and real-time motor control (H-bridge driver circuit)
- Transitioned from the breadboard prototype into my own custom 2-layer PCBs in KiCad (vehicle board and controller board), then hand-soldered all components.

ORGANIZATIONS & LEADERSHIP

- **Applied Robotics Engineering (ARE) Club, Founder & President:** Founded CMU's first dedicated robotics engineering club; assembled 7-person founding board with two faculty advisors; developing mechatronics and autonomous systems curriculum. (Jan 2026-Present)
- **Drone Club, Treasurer**: Lead collaborative drone design projects and manage club budget (August 2024 - Present)
- **Institute of Electrical and Electronics Engineers, Project Manager:** Organize workshops on PCB design and embedded systems; oversee student project development (August 2024 – Present)
- **CSE Campus Ambassador** — Lead technical facility tours for 100+ prospective students for the College of Science and Engineering; facilitate faculty-student engagement. (Aug 2025 – Present)
- **Student Coordinator, Student Food Pantry** — Coordinate weekly food distribution events for a university food resource program serving 400+ students in need (Jan 2024 – Present)